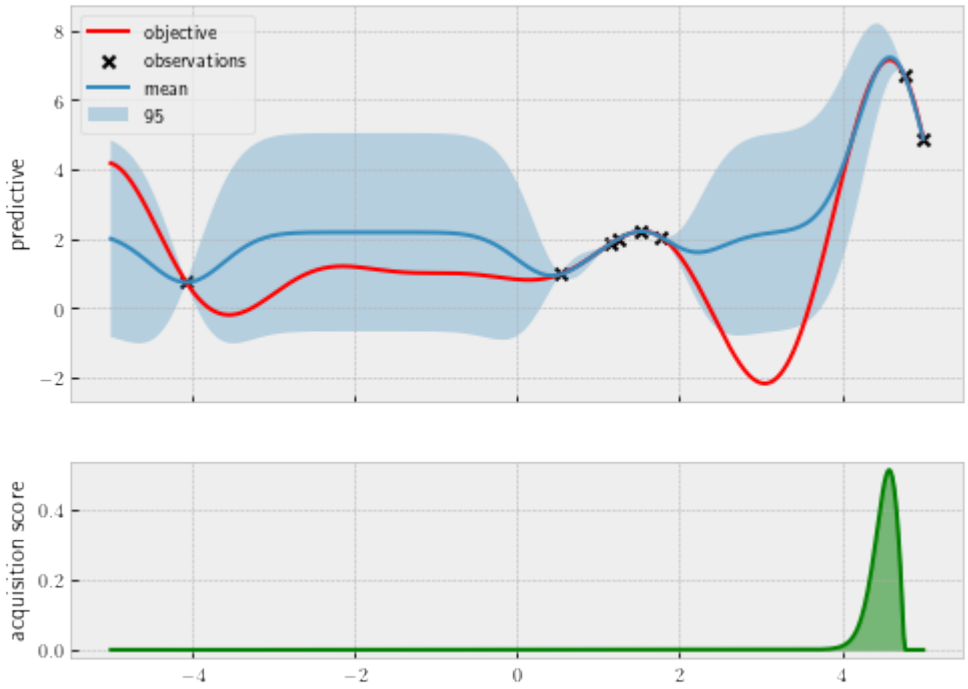


Figure 1.12 Bayesian optimization successfully ignores the large region on the left.

Here, we have observed two more points on the right, which updates both our GP model and our policy. Looking at the mean function (the solid line, representing the most likely prediction), we see that it almost matches with the true objective function from 4 to 5. Further, our policy (the green curve) is now pointing very close to the global optimum and basically no other area. This is interesting because we have not thoroughly inspected the area on the left (we only have one observation to the left of 0), but our model believes that regardless of how the function looks like in that area, it is not worth investigating compared to the current region. This is in fact true in our case here.

Finally, at the end of the search with ten queries, our workflow is now visualized in Figure 1.13. There is now little doubt that we have identified the global optimum around 4.3.

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